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THERMACOND, a code to compute lattice thermal conductivity from harmonic and anharmonic force constants

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THERMACOND is an open-source software for ab initio calculations of phonon thermal transport in crystalline bulk materials. It optimizes computational efficiency by leveraging the crystal symmetry to solve the linearized phonon Boltzmann transport equation (LPBTE) over the irreducible wedge of the Brillouin zone (IBZ). Brillouin zone integrations of δ functions are handled using the tetrahedron method. The code is written in Fortran90, parallelized using MPI to manage the computational costs associated with large k-point meshes, and is distributed under the GNU public license GPLv3. Here we provide an overview of the program structure and present results for three examples: Germanium (Ge), Germanium Selenide (GeSe), and diamond.

Heat conduction in semiconductors and insulators, driven by the quanta of lattice vibrations known as phonons, is a fundamental process that not only provides deep physical insights into various phenomena but also plays a crucial role in numerous energy technologies such as thermoelectricity¹, heat management, energy storage², high-power electronics, non-volatile memory based on phase-change materials³, and thermal medical devices⁴.

A fast and robust predictive theory for evaluating thermal conductivity that easily interfaces with modern ab initio codes has paved the way to an efficient and accurate modeling of heat transport phenomena in solids and devices. Semi-empirical models such as Slack's⁵ formula, originally proposed by Liebfried and Schlomann⁶ while computationally inexpensive, rely on experimental data. Likewise, ab initio molecular dynamics (MD) combined with Green-Kubo theory^{7,8} depends on force fields, though recent advancements in machine learning force fields trained on ab initio datasets have significantly improved their accuracy. However, it can be time-consuming as the MD simulations are done for one temperature and need to be ensemble averaged. Accurate thermal transport predictions require a microscopic understanding of phonon transport in solids, particularly insights into phonon dispersion and both intrinsic and extrinsic scattering mechanisms. In 1929, Peierls⁹ adapted Boltzmann's theory¹⁰ for gas transport to lattice vibrations, describing how phonon distribution functions deviate from thermal equilibrium under an applied temperature gradient. He also identified intrinsic thermal resistance using the lowest-order theory of phonon-phonon interactions, which includes normal and Umklapp processes that return phonons to equilibrium, thereby determining phonon lifetimes and consequently, thermal conductivity^{11,12}.

The linearized Boltzmann transport equation is commonly solved either approximately within relaxation time approximation or exactly, using either empirical or first-principles interactions, through iterative^{13,14}, variational^{15–17}, or direct diagonalization methods^{18,19}. These calculations have recently been implemented in various computational packages, including ALAMODE²⁰, almaBTE²¹, PhonTS²², HiPhive²³, Phono3py²⁴, and ShengBTE²⁵. Over the past decade, these approaches have been successfully applied to many systems, with good agreement with experimental results highlighted in numerous studies^{26–29}.

The iterative method for solving the LPBTE exactly was first introduced by Omini and Sparavigna¹³. As noted by these authors, there is no general proof guaranteeing convergence with this approach. It requires an extremely dense grid to achieve convergence in thermal conductivity. To address this limitation, several alternative methods have been developed recently. For instance, Fugallo et al.¹⁵ developed a stable algorithm using the variational principle and the conjugate gradient method, which is faster than the iterative procedure and consistently converges to the exact solution. Cepellotti et al.¹⁹ demonstrated that the eigenvectors of the scattering matrix in the Boltzmann equation correspond to collective phonon excitations known as relaxons. These excitations have well-defined relaxation times and provide an exact description of thermal transport through a kinetic theory of the relaxon gas. However, they also noted that this method may not be ideal for rapid calculations of transport properties³⁰.

This paper presents similar approaches for solving the LPBTE: iterative and direct (noniterative). In both cases, crystal symmetry is leveraged to reduce the computational time needed to achieve a satisfactory level of convergence by solving the LPBTE over the irreducible wedge of the

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Brillouin zone rather than the entire first Brillouin zone. As mentioned earlier, due to the numerical instabilities associated with the iterative approach, and to enhance calculation efficiency, we present the direct noniterative approach, which has none of the related numerical instabilities. To the best of our knowledge, this approach has also been recently implemented in Phoebe³⁰. This approach makes the THERMACOND code suitable for rapid calculations of phonon thermal transport properties. This article provides an overview of the program's structure and presents THERMACOND's capabilities, illustrated with the calculation of the phonon transport properties of Ge, GeSe, and diamond. As one of the most thermally conductive materials, diamond serves as an ideal benchmark for assessing predictive theoretical models of phonon thermal transport. Its relatively weak phonon-phonon Umklapp scattering plays a crucial role in the differences between the relaxation time approximation and full solution of the linearized phonon Boltzmann transport equation. The agreement between THERMACOND results, prior theoretical and experimental studies, and results from other available computational packages further validates the methodologies used in this work and confirms the reliability of the obtained results.

Results and discussion

This section presents the results of phonon properties, scattering mechanisms, and thermal conductivity for Germanium (Ge), Germanium Selenide (GeSe), and diamond. It begins with a description of key phonon properties and the phonon Boltzmann transport equation, along with the various solution methods implemented in THERMACOND. This is followed by details of the code's implementation, workflow, and input/output structure. Finally, a detailed analysis of the results is presented, including phonon linewidths derived from the imaginary part of the three-phonon self-energy and the thermal conductivities for Ge, GeSe, and diamond. The reliability of the results is validated by good agreement with experimental data and benchmark computational packages such as ShengBTE.

Phonon properties

Within the Born-Oppenheimer (BO) or adiabatic approximation, the lattice potential energy is described by a Taylor expansion around its equilibrium state.

$$V = V_0 + \sum_{\mathbf{R}\tau\alpha} \Pi_{\mathbf{R}\tau\alpha}^\alpha u_{\mathbf{R}\tau}^\alpha + \frac{1}{2} \sum_{\mathbf{R}\tau\alpha} \sum_{\mathbf{R}'\tau'\beta} \Phi_{\mathbf{R}\tau,\mathbf{R}'\tau'}^{\alpha\beta} u_{\mathbf{R}\tau}^\alpha u_{\mathbf{R}'\tau'}^\beta + \frac{1}{3!} \sum_{\mathbf{R}\tau\alpha} \sum_{\mathbf{R}'\tau'\beta} \sum_{\mathbf{R}''\tau''\gamma} \Psi_{\mathbf{R}\tau,\mathbf{R}'\tau',\mathbf{R}''\tau''}^{\alpha\beta\gamma} u_{\mathbf{R}\tau}^\alpha u_{\mathbf{R}'\tau'}^\beta u_{\mathbf{R}''\tau''}^\gamma + \dots \quad (1)$$

where $u_{\mathbf{R}\tau}^\alpha$ is the atomic displacements with $\mathbf{R}$ as a translation lattice vector of the (primitive) unit cell, τ is an index labeling the atoms in the unit cell, and α, β, γ are the Cartesian directions of the atomic displacement. Φ and Ψ represent the second and third derivatives of the potential energy with respect to atomic displacements and are known as the harmonic and cubic force constants, respectively. In the second term in the expansion, Π equals the negative of the net force acting on atom τ in the α direction, which is zero if the reference configuration is the equilibrium one minimizing the potential energy with respect to atomic displacements.

$$\begin{aligned} \Phi_{\mathbf{R}\tau,\mathbf{R}'\tau'}^{\alpha\beta} &= -\frac{\partial^2 V}{\partial u_{\mathbf{R}\tau}^\alpha \partial u_{\mathbf{R}'\tau'}^\beta} \\ \Psi_{\mathbf{R}\tau,\mathbf{R}'\tau',\mathbf{R}''\tau''}^{\alpha\beta\gamma} &= -\frac{\partial^3 V}{\partial u_{\mathbf{R}\tau}^\alpha \partial u_{\mathbf{R}'\tau'}^\beta \partial u_{\mathbf{R}''\tau''}^\gamma} \end{aligned} \quad (2)$$

These quantities are computed using FOCES which interfaces with DFT calculations in a supercell using VASP^{31,32}, Quantum-ESPRESSO³³, SIESTA^{34,35} or any similar package. They are used to characterize phonon properties and their scattering mechanisms. For instance, the harmonic properties such as phonon frequencies and eigenvectors are found by

diagonalizing the dynamical matrix

$$\sum_{\tau'\beta} D_{\tau\tau'}^{\alpha\beta}(\mathbf{k}) e_k^{\beta\tau'} = \omega_k^2 e_k^{\alpha\tau} \quad (3)$$

where the dynamical matrix D is the mass-scaled Fourier transform of the second derivative, Φ , of the potential energy in real space (where m_τ is the mass of atom τ in the primitive unit cell). ω_k is the phonon frequency, and a phonon state $k = (\lambda, \mathbf{k})$ consists of the phonon mode or branch index λ and the wave vector $\mathbf{k}$ in the first Brillouin zone. $e_k^{\alpha\tau}$ denotes the phonon eigenvector corresponding to atom τ and Cartesian direction α .

$$D_{\tau\tau'}^{\alpha\beta}(\mathbf{k}) = \sum_{\mathbf{R}} \frac{\Phi_{0\tau,\mathbf{R}\tau'}^{\alpha\beta}}{\sqrt{m_\tau m_{\tau'}}} e^{i\mathbf{k}\cdot\mathbf{R}} \quad (4)$$

From phonon dispersions, the phase space of all possible three-phonon scattering processes, which conserve both energy and crystal momentum, $\mathbf{k} + \mathbf{k}' + \mathbf{k}'' = \mathbf{G}$, due to translational invariance of the potential energy, is determined. Here, $\mathbf{G}$ represents a reciprocal lattice vector. Phonon-phonon interaction processes with $\mathbf{G} = 0$ and $\mathbf{G} \neq 0$ are called Normal and Umklapp processes, respectively.

From these quantities, particularly the lowest-order anharmonic term of the potential energy, various anharmonic material properties can be calculated, including mode Grüneisen parameters and thermal conductivity. The Grüneisen parameters, proportional to the thermal expansion, describe the phonon mode frequency dependence on crystal volume, are defined by $\gamma_k = -\frac{d\ln(\omega_k)}{d\ln(V)}$. It is a good indication of anharmonicity strength, and is related to the third derivative of the potential energy Ψ through the following Eq.³⁶:

$$\gamma_k = \frac{-1}{6\omega_k^2} \sum_{\tau} \sum_{\mathbf{R}\tau'} \sum_{\mathbf{R}'\tau''} \sum_{\alpha\beta\gamma} \Psi_{0\tau,\mathbf{R}\tau',\mathbf{R}'\tau''}^{\alpha\beta\gamma} \frac{e^{i\mathbf{k}\cdot(\mathbf{R}-\mathbf{R}')}}{\sqrt{m_\tau m_{\tau'}}} \times X_{0\tau}^\alpha e_k^{\beta\tau'} e_k^{\gamma\tau''} \quad (5)$$

where $X_{\mathbf{R}\tau}$ denotes the equilibrium atomic position of atom τ in the primitive cell indexed by the translation vector $\mathbf{R}$. In low-symmetry crystals with multiple atoms per unit cell, an applied homogeneous strain can induce additional internal displacements^{37,38}, and this effect adds a small correction to $X_{0\tau}$. Although a more comprehensive approach would account for both external and internal contributions, our implementation focuses on the anharmonic term derived from third-order IFCs.

The phonon frequency shift and broadening (inverse lifetime) are determined by the real and imaginary parts of the three-phonon self-energy ($\Sigma = \Sigma' + i\Sigma''$) evaluated at the on-shell phonon frequency ω_k , Eq. (6)^{39–41}. These quantities are observable in Raman-scattering experiments, where the linewidth of the Raman line corresponds to $2\Sigma''$, and Σ' indicates the shift in the position of the line center due to anharmonicity. Σ' and the lifetime, $\tau_k = 1/2\Sigma''(\omega_k)$ depend on temperature T solely through the phonon distribution function n_k .

$$\begin{aligned} \Sigma(k, \omega) &= -\frac{1}{2} \sum_{k', k''} \sum_{s=\pm 1} |V_3(k', k'', k)|^2 \\ &\times \left[\frac{(n_{k'}^0 + n_{k''}^0 + 1)}{\omega_{k'} + \omega_{k''} + s\omega_c} + \frac{n_{k'}^0 - n_{k''}^0}{\omega_{k'} - \omega_{k''} + s\omega_c} \right] \end{aligned} \quad (6)$$

where $\omega_c = \omega - i\eta$, and $\eta \simeq 0^+$ is a small infinitesimal number, practically taken to be finite, on the order of $v_{\text{sound}}\Delta k$ for a given $\mathbf{k}$ -point mesh size Δk , n^0 is the equilibrium Bose-Einstein distribution function. The three-phonon matrix element V_3 , expressed as a function of the cubic interatomic force constants, is given by:

$$\begin{aligned} V_3(k', k'', k) &= \left(\frac{\hbar}{8N\omega_k\omega_{k'}\omega_{k''}} \right)^{1/2} \sum_{\tau\tau'} \sum_{\alpha\beta\gamma} \Psi_{0\tau,\mathbf{R}\tau',\mathbf{R}''\tau''}^{\alpha\beta\gamma} \\ &\times \frac{e^{i(\mathbf{k}'\cdot\mathbf{R} + \mathbf{k}''\cdot\mathbf{R}'')} e_k^{\alpha\tau} e_{k'}^{\beta\tau'} e_{k''}^{\gamma\tau''}}{\sqrt{m_\tau m_{\tau'} m_{\tau''}}} \end{aligned} \quad (7)$$

Given the identity $(\frac{1}{x \pm i0^+} = \mathcal{P}(\frac{1}{x}) \mp i\pi\delta(x))$ the imaginary part of the phonon self-energy Σ'' involves δ functions reflecting the conservation of energy of the three phonon processes, $\omega_k = \omega_{k'} \pm \omega_{k''}$. In practice, the latter is substituted by a Lorentzian function of finite width η in Eq. (6). A more accurate alternative way to perform the integration including the Dirac delta functions is to use the tetrahedron method^{42,43}. This method eliminates the need for the broadening parameter η , making the results more reliable and convergent with a coarser k-mesh, thereby ensuring greater accuracy in the calculations. The δ functions arising from energy conservation in the next section are calculated using this method. The frequency shifts are obtained from the solution of $\omega = \omega_k + \Sigma'(k, \omega)$, which, practically, can be obtained by substituting the “on-shell” frequency in the right-hand side: $\omega \simeq \omega_k + \Sigma'(k, \omega_k)$.

Here, the “on-shell” frequency refers to the phonon frequency obtained within the harmonic approximation, using the second-order interatomic force constants. The real and imaginary parts of the self-energy ($\Sigma = \Sigma' + i\Sigma''$), which are functions of frequency ω , are evaluated at the harmonic phonon frequency ($\omega = \omega_k$), rather than a renormalized or shifted frequency. This simplifies the calculation of frequency shifts and broadenings, and avoids a self-consistent solution.

Phonon Boltzmann transport equation (PBTE)

The lattice thermal conductivity is calculated from the solution to the phonon Boltzmann transport equation (PBTE) and the microscopic quantities discussed in the previous section. The Boltzmann transport equation describes the time evolution of the phonon distribution function in the presence of a temperature gradient across a dielectric solid. It accounts for two mechanisms that contribute to the total rate of change of the distribution: diffusion (the first term in Eq. (8) containing ∇_r describing the phonon distribution change in the presence of a temperature gradient) and scattering (the second term, also called the collision integral). In this work, to obtain the lattice thermal conductivity, we solve the steady-state PBTE where the partial time derivative (left hand side) is set to zero.

$$\frac{\partial n_k}{\partial t} = -\mathbf{v}_k \cdot \nabla n_k + \frac{\partial n_k}{\partial t}_{\text{coll}} \quad (8)$$

where $\mathbf{v}_k$ is the phonon group velocity of the state $k = (\lambda, \mathbf{k})$. In the steady state of heat flow through the solid, the total rate of change of the distribution must vanish. In the presence of a sufficiently small thermal gradient ∇T , the phonon distribution function can be expanded linearly around the equilibrium distribution, $n_k = n_k^0 + n_k^1$, where n_k^0 is the equilibrium distribution and n_k^1 is the perturbation due to the thermal gradient, leading to the following linearized PBTE:

$$-\mathbf{v}_k \cdot \nabla T \frac{\partial n_k^0}{\partial T} = -\frac{\partial n_k^1}{\partial t}_{\text{coll}} = \sum_{k'} C_{kk'} n_k^1 \quad (9)$$

The collision term accounts for the relaxation of phonons due to various scattering processes, including intrinsic anharmonic scattering and extrinsic processes such as impurity scattering. The collision matrix elements, $C_{kk'}$, are obtained using second-order perturbation theory, where the transition probabilities are linearized by $n_k^1 = n_k^0(n_k^0 + 1)\psi_k$ ¹¹. Here, ψ_k represents the unknown measure of the deviation from the equilibrium distribution for phonons in state k , which can be determined by solving the linearized PBTE.

$$\begin{aligned} \sum_{k'} C_{kk'} n_k^1 &= \sum_{k'k''} [P_{kk'}^{k''}(\psi_k + \psi_{k'} - \psi_{k''}) \\ &\quad + \frac{1}{2} P_k^{k'k''}(\psi_k - \psi_{k'} - \psi_{k''})] \\ &\quad + \sum_{k'} P_k^{k'}(\psi_k - \psi_{k'}) \end{aligned} \quad (10)$$

The intrinsic collision rates for coalescence processes ($P_{kk'}^{k''}$), and decay processes ($P_k^{k'k''}$), obtained from the Fermi golden rule (FGR), and the mass

disorder scattering rate ($P_k^{k'}$) are given by

$$\begin{aligned} P_{kk'}^{k''} &= 2\pi n_k^0(n_{k'}^0 + 1)|V_3(-k, -k', k'')|^2 \delta(\omega_k + \omega_{k'} - \omega_{k''}) \\ P_k^{k'k''} &= 2\pi n_k^0(n_{k'}^0 + 1)(n_{k''}^0 + 1)|V_3(-k, k', k'')|^2 \delta(\omega_k - \omega_{k'} - \omega_{k''}) \\ P_k^{k'} &= \frac{\pi}{2} \omega_k \omega_{k'} n_k^0(n_{k'}^0 + 1) \left(\sum_{\tau} g_{\tau} |\mathbf{e}_{\tau k}^* \cdot \mathbf{e}_{\tau k'}|^2 \right) \delta(\omega_k - \omega_{k'}) \end{aligned} \quad (11)$$

where the δ function, computed using the tetrahedron method, ensures energy conservation between the initial and final phonon states, and V_3 are the three-phonon anharmonic matrix elements, related to the third-order interatomic force constants in Eq. (7). The mass-variance (or point defect scattering) parameter g of atom type τ is defined by $g_{\tau} = \sum_i f_{\tau,i} (1 - \frac{M_{\tau,i}}{M_{\tau,avg}})^2$, where $f_{\tau,i}$ is the fraction of isotope i of atom τ in the primitive cell with $\sum_i f_{\tau,i} = 1$. A similar term can also be considered for bond disorder if the latter is strong. It requires the knowledge of bond strengths around the impurity.

As the simplest approximation, in the relaxation time approximation (RTA), it is assumed that all the phonon modes except for mode k are in thermal equilibrium with occupation n^0 (i.e., $\psi_k \neq 0$, but $\psi_{k'} = \psi_{k''} = 0$). This leads to a diagonal collision matrix, which is easily invertible. The phonon lifetime can then be obtained from the diagonal collision matrix ($\frac{\partial n_k^1}{\partial t}_{\text{coll}} = \frac{n_k - n_k^0}{\tau_k} = Q_k \psi_k$), where Q_k and τ_k are given by

$$Q_k = \sum_{k'k''} \left(P_{kk'}^{k''} + \frac{1}{2} P_k^{k'k''} \right) + \sum_{k'} P_k^{k'} \quad (12)$$

$$\tau_k = \frac{n_k^0(n_k^0 + 1)}{Q_k} \quad (13)$$

The thermal conductivity can then be calculated using the following relation:

$$\kappa_{\alpha\beta} = \frac{1}{\Omega N_k} \sum_k \tau_k v_k^{\alpha} v_k^{\beta} C_k \quad (14)$$

where $C_k = \hbar \omega_k \frac{\partial n_k^0}{\partial T}$ is the heat capacity per mode, Ω is the unit cell volume, and N_k is the number of k-points in the summation. The RTA is accurate when Umklapp+impurity or dissipative scattering predominates over normal scattering.

LPBTE solvers

Iterative method. In this approach, implemented in the THERMA-COND code, the linearized phonon Boltzmann Transport Equation, Eq. (9), is solved iteratively. By defining $\psi_k = \sum_{\alpha} F_{k\alpha} \frac{\partial T}{\partial x_{\alpha}}$ and substituting it into Eq. (10), the LPBTE, Eq. (9), can be rewritten as Eq. (17) and solved iteratively. To reduce the computation time, the symmetry of the crystal is exploited to solve the LPBTE over the irreducible wedge of the Brillouin zone (IBZ) rather than the whole first Brillouin zone (FBZ). The complete set of a general vector, such as the group velocity $\mathbf{v}_k$ or the out-of-equilibrium phonon population $\mathbf{F}_k$ in the FBZ, is obtained by applying point group symmetry operation matrices $\mathcal{S}_k$ to each of their corresponding vector, $\mathbf{v}_{k^*}$ or $\mathbf{F}_{k^*}$, with k^* in the IBZ. They are transformed in the same way as the wave vectors, $\mathbf{k} = \mathcal{S}_k \mathbf{k}^*$.

$$\mathbf{v}_k^{\alpha} = \sum_{\beta} \mathcal{S}_k^{\alpha\beta} \mathbf{v}_{k^*}^{\beta} \quad (15)$$

$$\mathbf{F}_k^{\alpha} = \sum_{\beta} \mathcal{S}_k^{\alpha\beta} \mathbf{F}_{k^*}^{\beta} \quad (16)$$

Thus the sums over wavevectors in the full zone are written as a double sum over their star in the irreducible zone and the symmetry operations which

unfold the irreducible wavevector onto the full zone:

$$F_{k^*}^{\alpha} = \frac{\hbar\omega_{k^*} n_{k^*}^0 (n_{k^*}^0 + 1) v_{k^*}^{\alpha}}{k_B T^2 Q_{k^*}} + \frac{1}{Q_{k^*}} \left\{ \sum_{k''} \sum_{S'} \left[P_{k''}^{k'} \left(S_{k''}^{\alpha\beta} F_{k''}^{\beta} - S_{k'}^{\alpha\beta} F_{k''}^{\beta} \right) + \frac{1}{2} P_{k''}^{k'} \left(S_{k''}^{\alpha\beta} F_{k''}^{\beta} + S_{k'}^{\alpha\beta} F_{k''}^{\beta} \right) \right] + \sum_{k''} \sum_{\beta} P_{k''}^{k'} S_{k''}^{\alpha\beta} F_{k''}^{\beta} \right\} \quad (17)$$

$$F_{k^*}^{0\alpha} = \frac{\hbar\omega_{k^*} n_{k^*}^0 (n_{k^*}^0 + 1) v_{k^*}^{\alpha}}{k_B T^2 Q_{k^*}} \quad (18)$$

Since the LPBTE (Eq. (17)^{13,14}) contains F_{k^*} on both sides, it is solved iteratively. The process begins with an initial guess, $F_{k^*}^0$, which is substituted into the right-hand side of the equation. The updated F_{k^*} is then computed and reinserted iteratively until the solution converges. Once convergence is achieved, the thermal conductivity tensor is determined using the following expression.

$$k_{\alpha\beta} = \frac{1}{N_0 \Omega} \sum_{k^*, S} \hbar\omega_{k^*} n_{k^*}^0 (n_{k^*}^0 + 1) S^{\alpha\alpha'} v_{k^*}^{\alpha'} S^{\beta\beta'} F_{k^*}^{\beta'} \quad (19)$$

The convergence criterion is that the relative change in the conductivity tensor, calculated using Eq. (19), is less than a customizable parameter, typically on the order of 0.001, but which can be read from the input-file parameters. Due to momentum conservation, Eq. (17) only involves a single summation over the complete set of wave vectors $\mathbf{k}$ in the full Brillouin zone. Using both Eq. (16) and Eq. (15), the summation is then restricted to the irreducible Brillouin zone. The scalar quantities such as the phonon frequencies ω_k , phonon distribution functions n_k^0 , and Q_k or lifetimes τ_k remain invariant under the application of symmetry operations. Setting $F_{k'}, F_{k''} = 0$ in Eq. (17) and only using $F_{k^*}^0$ is equivalent calculating the thermal conductivity within the relaxation time approximation.

Direct method. In the iterative procedure, as indicated by the authors in ref. 15, there is no general proof that convergence will always be achieved. It usually requires a fine mesh of wave vectors. Even with the use of crystal symmetry to compute the summation in Eq. (17) over the irreducible zone, as discussed in the previous section, it still demands a substantial amount of computing time. Here, we present a direct noniterative solution to the Boltzmann transport equation. In this method, the crystal symmetry is used to reduce memory usage and further accelerate the simulations. By multiplying the collision matrix C and the crystal symmetry matrix S , a square matrix M of order n is generated, where n is the number of k -points in the irreducible wedge of the Brillouin zone, times the number of modes. Eq. (17) is then given by the following expression

$$v_{k^*} \frac{\partial n_{k^*}^0}{\partial T} = \sum_{k''} M_{k^* k''} F_{k''} ; M_{k^* k''} = \sum_{k'} C_{k^* k''} S_{k'' k'} \quad (20)$$

$$C_{kk_1} = \left(\sum_{k''} \left(-P_{kk''}^{k'} - \frac{1}{2} P_k^{k' k''} \right) + \sum_{k'} -P_k^{k'} \right) \delta_{k, k_1} + \left(\sum_{k''} \left(-P_{k, k_1}^{k'} + P_{kk''}^{k_1} + P_k^{k_1 k''} \right) + P_k^{k_1} \right) (1 - \delta_{k, k_1}) \quad (21)$$

The matrix M is sufficiently small to be solved numerically using either the singular value decomposition (SVD) approach or the conjugate gradient method. The accuracy is primarily controlled by the number of k -points used to sample the first Brillouin zone. After extracting F_{k^*} from Eq. (20), the thermal conductivity tensor is computed using Eq. (19).

Implementation

THERMACOND is designed to efficiently solve the linearized phonon Boltzmann transport equation while addressing key computational challenges related to convergence, memory usage, and efficiency. The code is written in Fortran90 and structured into modular components to facilitate extensibility, numerical stability, and high-performance calculations. Three methods are implemented for solving the LPBTE: (1) In the relaxation time approximation, the diagonal elements of the collision matrix are used to estimate phonon lifetimes and thermal conductivity. While computationally efficient, the RTA does not properly take into account momentum-conserving normal processes, often underestimating the thermal conductivity. (2) The iterative approach updates the out-of-equilibrium phonon population in successive steps until convergence. This approach avoids explicit storage of the full collision matrix but lacks a general proof of convergence. To enhance efficiency, THERMACOND exploits crystal symmetry by restricting calculations to the irreducible wedge of the Brillouin zone, reducing computational cost. However, achieving convergence often requires dense k -point sampling, making this approach computationally demanding. (3) To circumvent the convergence limitations of the iterative method, THERMACOND implements a direct approach by constructing a reduced matrix, M , obtained by projecting the full collision matrix onto the IBZ using point group symmetry operations. This matrix is then solved using singular value decomposition or conjugate gradient methods.

While the iterative method is implemented, THERMACOND primarily relies on the direct approach, as it achieves identical results with significantly faster computational time.

Instead of conventional Gaussian or Lorentzian broadening techniques, THERMACOND employs the tetrahedron method, ensuring the accuracy of Brillouin zone integrations of δ functions arising from the conservation of energy. This improves the accuracy of the phonon lifetime and enhances the overall reliability of thermal transport predictions.

THERMACOND is parallelized using MPI, enabling efficient scaling across large supercomputing clusters. The parallelization is designed to distribute k -points across processors, and the symmetry reduction of the k -point grid in the IBZ further minimizes the computational burden, ensuring that THERMACOND can handle large systems with high efficiency.

General structure of THERMACOND

The workflow for thermal transport calculation, starting with the determination of the relaxed crystal structure is shown in Fig. 1. The steps involved in THERMACOND are outlined below:

1. The unit cell parameters and equilibrium atomic positions can be obtained from an energy minimization in DFT.
2. The harmonic, cubic, and quartic force constants are calculated from FOCX.
3. Harmonic lattice dynamics calculations, by diagonalizing the dynamical matrix, determine the set of phonon frequencies and eigenvectors corresponding to the wave vector $\mathbf{k}$. This provides the phonon dispersion curves, from which the phonon group velocities, phonon density of states, and Bose-Einstein distribution functions are computed.
4. Gruneisen parameters are calculated using the eigenvectors and eigenvalues of the dynamical matrix and the cubic force constants from FOCX.
5. The phonon frequency shifts and linewidths are calculated as the real and imaginary parts of the three-phonon self-energy, computed using the eigenvectors and eigenvalues of the dynamical matrix and the cubic force constants.
6. The cubic interatomic force constants are used to compute the diagonal collision matrix, which, along with the phonon frequencies and equilibrium phonon distribution functions, yield the phonon relaxation times and thermal conductivity in the RTA approximation.
7. The full collision matrix, computed using three-phonon matrix elements and equilibrium phonon distribution functions, are used to

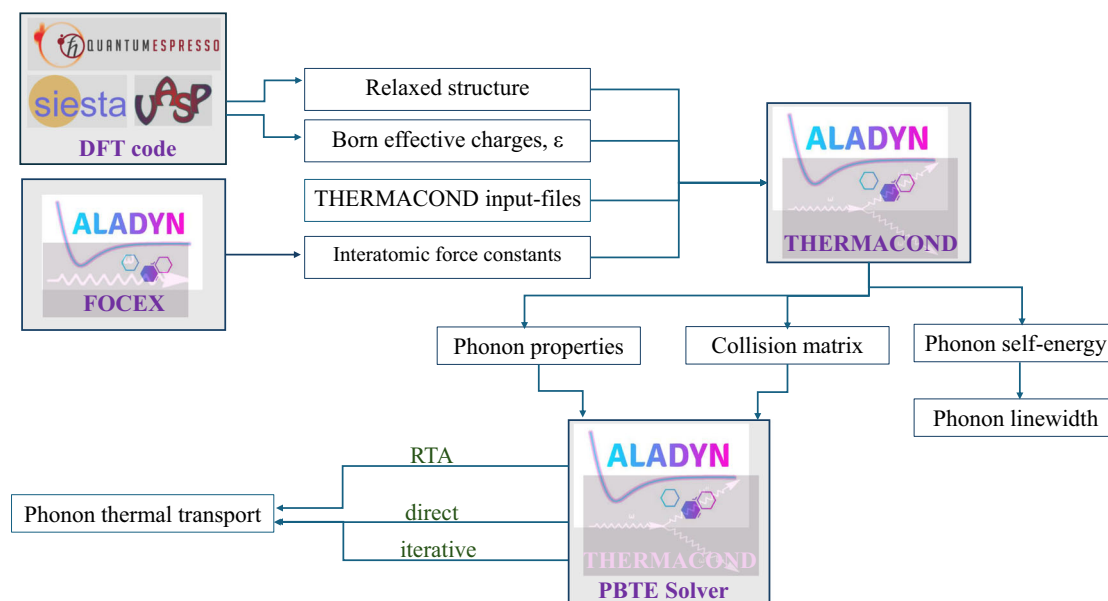


Fig. 1 | Workflow of THERMACOND for phonon thermal transport calculations. DFT codes (Quantum ESPRESSO, VASP, SIESTA) provide the relaxed crystal structure and Born effective charges, which are used by the FOCEX to generate interatomic force constants and prepare THERMACOND input files.

THERMACOND first calculates the phonon properties, collision matrix, and phonon self-energy. These outputs are passed to its LPBTE solver to compute the lattice thermal conductivity using the relaxation time approximation (RTA), direct, or iterative solution of the linearized phonon Boltzmann transport equation.

solve the LPBTE over the irreducible k-points using either an iterative or direct approach to obtain the non-equilibrium phonon distribution function, and subsequently, the thermal conductivity.

Input and output files

THERMACOND requires the following set of input files to run. Full details about their formats can be found in the example on GitHub.

- `<lat_fc.dat>`, generated by FOCEX, contains information about irreducible and total number of FCs, and the coordinates and identity of the atoms within and in the neighborhood of the primitive cell.
- `<fc2.dat>`, `<fc3.dat>` are harmonic and cubic interatomic force constants also generated by FOCEX.
- `<latdyn.params>` specifies a set of parameters and flags controlling the execution of THERMACOND. This includes the k-point mesh, smearing, temperature window, whether or not to run MPI, isotope scattering... An example of this input file is shown below.


```
14 14 14 #k-mesh for Brillouin zone integration
0 0 0 1 0 0 #shift in k-mesh units; normal to the
surface (optical properties)
300 350 #frequency mesh, max freq (1/cm) for DOS
10 30 #broadening width for DOS, eta=I-
m(om_c) eq.6
.False. #verbosity
0 900 41 #T_initial (K), T_final (K), # of T values
0 #0=quantum; 1= classical distribution function
0 #0=bands & DOS only; 1= MPI for V3 & k
#2= linewidths; 3< for testing
2 #ksub_size=#kpoints per CPU for splitjob
./ #path where output files will be written
0 #1=include isotope scattering
0 0 0 0 #calc_cross at q_cross
100000 #sample length in microns
```
- `<structure.params>` is the file used by FOCEX to generate IFCs. But it is needed for reading the lattice parameters and the number of IFC neighbor shells. An example of this input file for Ge is shown below.

```
1 1 1 90 90 90 #a,b,c,alpha,beta,gamma
conventional
0 0.5 0.5, 0.5 0 0.5, 0.5 0.5 0 #conv to prim mtrx
5.7022565 #scaling factor of a,b,c
1 1 1 0 0 0 0 0 #1 to include FCs of rank 1,...,8
1 1 0 0 0 #translational,rotational,Huang
invariances
0 300 #1=weight snapshots by Boltzmann
factor,T(K)
1 .false. #number of FORCEDISP files, verbosity
1 #number of atom_types
72.64 #atomic masses
Ge #names of elements
2 #number of atoms in the primitive unit cell
0 #0: default range of FC2s; 1: use range below
9 9 #number of shells for FCs rank 2 for each atom
6 6 #number of shells for FCs rank 3 for each atom
0 0 #number of shells for FCs rank 4 for each atom
0 0 #number of shells for FCs rank 5 for each atom
0 0 #number of shells for FCs rank 6 for each atom
0 0 #number of shells for FCs rank 7 for each atom
0 0 #number of shells for FCs rank 8 for each atom
1 1 0 0 0 #atom number, atom_type, position (con-
ventional cell units)
2 1 0.25 0.25 0.25 #atom number, atom_type,
position
```

In this study, only second- and third-order interatomic force constants were used to compute the phonon properties, collision matrix, and self-energy. The input file format demonstrates THERMACOND's capability to incorporate higher-order IFCs, including fourth-order and beyond, by specifying the number of shells for each rank. However, to maintain consistency with the presented results, the fourth-order IFCs were explicitly set to zero in this work. Future extensions of THERMACOND will aim to include higher-order anharmonic interactions, such as four-phonon

scattering and phonon frequency renormalization, to further enhance the accuracy of thermal transport predictions.

- `<dielectric.params>` contains information about the dielectric constant and Born effective charges of the atoms in the primitive cell in the order they appear in `<structure.params>`. This file is necessary to account for the non-analytical part of the dynamical matrix due to the long-range electrostatic correction. The first line is a flag to account for them (0 to exclude), followed by three lines for the dielectric constant tensor, and followed by three lines for each Born effective charge.
- `<kpbs.params>` contains high symmetry k-point paths chosen by the user to calculate phonon dispersion curves. The first line specifies the unit of the k-points (1 for primitive or 0 for conventional reciprocal lattice), followed by the number of k-points along each direction, followed by the number of directions. In the following lines the label of each starting kpoint and its reduced coordinates appear. Last line has the label of the last k-point and its fractional coordinates.

```
0 #1 for primitive and 0 for conventional units
40 # number of kpoints along each direction
5 # number of directions
G 0 0 0 # starting k-point (name, coordinates)
X 0.5 0 0 # end of first direction
W 0.5 0.25 0 # end of 2nd direction
K 0.375 0.375 0
L 0.5 0.5 0.5
G 0 0 0 # last k-point; end of 5th direction
```

After running THERMACOND in the directory containing all the aforementioned input files, the following outputs will be generated:

- `<KPOINT.FBZ>` lists k-points and their weights in FBZ in cartesian and direct coordinates. It also contains mapping integer arrays for k-points in the FBZ and IBZ.
 - `<KPOINT.IBZ>` lists k-points and their weights in IBZ in cartesian and direct coordinates.
 - `<KPOINT.MP>` lists all k-points and their weights.
 - `<bs_freqs.dat>` phonon dispersion curves along the high symmetry k-point paths chosen by the user (second column vs the $3 \times N_{atoms}$ columns starting from the sixth column). The first five columns contain information about k-points.
 - `<dos_tet.dat>` phonon density of states using the tetrahedron method (phonon frequencies in the first column in cm^{-1} , total DOS and integrated DOS in the 3rd and 4th columns; dos per band in the remaining columns).
 - `<veloc.dat>` contains group velocities at each k-point in the whole Brillouin zone and for each phonon band.
 - `<bs_grun.dat>` Grüneisen parameter along the high symmetry k-point paths chosen by the user (second column) for each phonon mode (the $3 \times N_{atoms}$ columns starting from the sixth column). The first five columns contain information about k-points.
 - `<kappa_RTA.dat>` thermal conductivity tensor from each band in units of (W/mK) in the RTA approximation.
 - `<kappa_direct-cg.dat>` thermal conductivity tensor from each band in units of (W/mK) using the direct approach.
 - `<kappa_iter.dat>` converged values of the tensorial contribution to the thermal conductivity from each band in units of (W/mK).
 - `<tau_RTA.dat>` total, Normal, Umklapp phonon lifetimes in the RTA approximation at each k-point in the irreducible wedge of the Brillouin zone and their respective phonon bands.
 - `<tau_RTAAandeff.dat>` also contains total phonon lifetimes, calculated using the non-equilibrium phonon distribution function, at each k-point in the irreducible wedge of the Brillouin zone and their respective phonon bands.
- `<Qvalue.dat>` the last two columns are Q and the three-phonon scattering rates, respectively, in the RTA approximation, for phonon

modes in the irreducible wedge of the Brillouin zone, in s^{-1} . The first two columns correspond to the indices of the k points numbered in the IBZ and their phonon bands, respectively.

- `<phonlog.dat>` contains log of the entire run.
- `<times.dat>` shows CPU times at each computational step.

Examples

Ge. In addition to its essential role as a pivotal element in modern technology, Ge serves as an exemplary material for investigating thermal transport mechanisms and testing theoretical models. Following the workflow presented in Fig. 1, the sets of IFCs were computed from FOCX using $3 \times 3 \times 3$ supercell (216-atom supercells) generated based on crystal symmetries.

36 snapshots in which only one or two atoms have been moved were generated. Forces were calculated using VASP^{44–47} with the GGA-PBESol functional, at cutoff energy of 225 eV and a Gamma-centered $7 \times 7 \times 7$ k-point mesh within the supercell. The lattice constant of $a = 5.702 \text{ \AA}$ was found using $14 \times 14 \times 14$ k-mesh for the primitive cell optimization calculations. This is only 0.7% larger than the experimental value of $5.66 \text{ \AA}$. All harmonic FCs within the supercell were included in the fitting, while cubic force constants were included up to the 6th nearest neighbors corresponding to a cutoff distance of $6.98 \text{ \AA}$. The slight discrepancy between the computed and experimentally measured⁴⁸ frequencies is most likely due to the larger PBE lattice constant of $5.7 \text{ \AA}$, which is slightly larger than the experimental value of $5.66 \text{ \AA}$. This is a 2.1% larger volume. Therefore for a Grüneisen parameter of +1 for instance, a 2.1% reduction in the phonon frequency should be expected.

Figure 2a, b shows the resulting germanium phonon dispersion along high-symmetry directions and the corresponding phonon density of states. Each phonon branch is represented by the respective intrinsic linewidth at 300 K. These linewidths were computed along high-symmetry directions by integrating the imaginary part of the three-phonon self-energy over a fine grid of $30 \times 30 \times 30$ k-points. Figure 2c also shows the Grüneisen parameters along the high-symmetry line in the Brillouin zone. There is a good agreement between the phonon dispersion and Grüneisen parameters with their experimentally reported results^{49–51}, illustrated by gray dots in Fig. 2a–c demonstrating strong consistency with experimental data and validating the methodologies used here. Figure 2e indicates the frequency dependence of the group velocity for each phonon mode. As expected, the acoustic modes have higher group velocities compared to the optical modes, predominantly contributing to thermal transport. Figure 2f also shows the total intrinsic phonon-phonon scattering rates (black), separated into normal scatterings (red) and Umklapp scatterings (blue). The strong Umklapp scattering implies that thermal conductivity is dominated by this scattering, making the relaxation time approximation result accurate, as shown in Fig. 2d. In this figure, the converged thermal conductivity computed directly is compared against experimental values^{52,53}, indicating strong agreement with both experimental data and previous theoretical studies. In the inset, to show the scaling versus the density of k-points in the first Brillouin zone, we have plotted different thermal conductivity values as a function of the inverse of the number of k-points along a direction. The densest mesh being 30^3 , the corresponding point in the horizontal axis is $1/30 = 0.033$. To get the converged value, the results have been extrapolated to $1/k \rightarrow 0$, and these are reported in the main plot (d).

GeSe. GeSe has been predicted to be a promising thermoelectric (TE) material given an optimal carrier concentration⁵⁴ or alloying with Ag-V-VI₂ compounds due to the formation of metavalent bonding (MVB)⁵⁵. It is well-known that the low-temperature phase of GeSe has an orthorhombic crystal structure with a *Pnma* space group.

Following the workflow presented in Fig. 1, the sets of IFCs were obtained from FOCX using 30 displaced $6 \times 5 \times 2$ supercells (480-

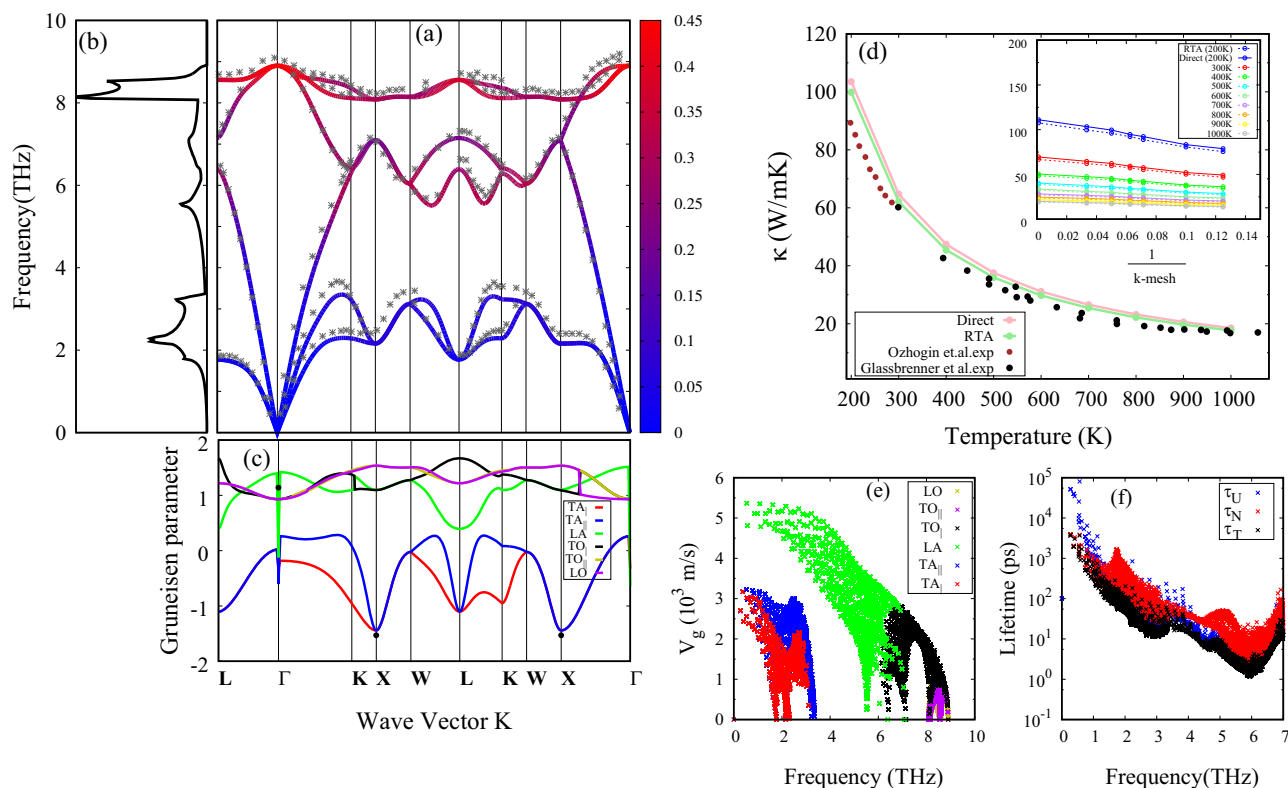


Fig. 2 | Phonon and phonon thermal transport properties of germanium.

a Phonon dispersion of germanium used in the thermal transport calculation. The color bar indicates the intrinsic phonon linewidth (imaginary part of the self-energy) in THz. Gray dots represent experimental phonon dispersion data from refs. 49,50), **b** Phonon density of states, **c** Mode Grüneisen parameters plotted along the high-symmetry lines in the Brillouin zone (black dots: experimental data from ref. 51),

d Thermal conductivity versus temperature. Brown and black dots represent experimental data from refs. 52,53. The inset shows the scaling versus the inverse of number of k-points along the x direction, **e** Group velocity of each phonon mode as a function of frequency, **f** Intrinsic three-phonon scattering rates (black), separated into normal scatterings (red) and Umklapp scatterings (blue).

atom supercells), considering shells for IFCs up to the 34 and 20 nearest neighbors, corresponding to cutoffs 7.53 Å and 6.1 Å, for harmonic, cubic, respectively. The supercell DFT calculations were performed using the SIESTA package^{34,35} and the Perdew-Burke-Ernzerhoff form of the generalized gradient approximation (PBE-GGA) for the exchange-correlation energy functional⁵⁶. A cutoff energy of 350 Ry for the real space integrals and a $3 \times 3 \times 3$ Monkhorst-Pack k-mesh were employed. The converged lattice parameters of $a = 3.83$ Å, $b = 4.39$ Å, and $c = 10.83$ Å, were obtained with the maximum forces on atoms reduced to less than 10^{-5} eV/Å, in good agreement with the experimental values $a = 3.83$ Å, $b = 4.39$ Å, and $c = 10.83$ Å^{57,58}. A $12 \times 10 \times 4$ k-point grid was then used for the THERMACOND run to calculate the phonon properties, linewidth, and thermal transport.

The phonon dispersions and the corresponding phonon density of states are shown in Fig. 3a, b, respectively, and are consistent with other theoretical calculations. The intrinsic phonon linewidths, computed over the same k-point grid at 300 K, are superimposed on the phonon dispersions. It is clear that the optical modes tend to scatter more, as the linewidths are a measure of the total scattering rate. Subsequently, the thermal conductivity was computed by directly solving the phonon Boltzmann transport equation. Figure 3 displays the lattice thermal conductivity as a function of temperature and compares the LPBTE results with experimental values⁵⁸. The good agreement with experimental data validates the results of this work. Figure 3 illustrates the frequency dependence of the lifetimes for both Normal and Umklapp processes for each mode. It is evident that the lifetimes are largest for the acoustic phonon modes with the highest group velocities, as shown in Fig. 3, which predominantly contribute to thermal transport.

Diamond. Diamond, the face-centered cubic (FCC) allotrope of carbon (space group $Fd\bar{3}m$), is an extraordinary material with a combination of high incompressibility, melting temperature, and thermal conductivity. It also exhibits an ultra-wide electronic bandgap and high charge carrier mobility⁵⁹, making it highly desirable for a variety of electronic and high-power applications. Among its many exceptional properties, diamond is one of highest thermally conductive materials, with values exceeding 3000 W/mK ^{60–62} at room temperature in isotopically enriched samples. This exceptionally high thermal conductivity arises from strong sp^3 covalent bonds and the low atomic mass of carbon, which result in high phonon frequencies and acoustic velocities. The phonon-phonon umklapp scattering around room temperature is unusually weak, enabling highly efficient thermal transport. These unique characteristics make diamond an ideal benchmark material for testing predictive theoretical models of phonon thermal transport.

Following the workflow presented in Fig. 1, the sets of IFCs were obtained from FOCX using 170 displaced $4 \times 4 \times 4$ supercells (128-atom supercells), considering shells for IFCs up to the 6th nearest neighbors, corresponding to a cutoff of 4.32 Å for cubic IFCs. All harmonic IFCs within the supercell were included in the fitting. The supercell DFT calculations were performed utilizing the Vienna ab initio simulation package (VASP)^{31,32} employing the projector augmented-wave (PAW) method⁶³ and the PBEsol form of the generalized gradient approximation (PBEsol-GGA) for the exchange-correlation energy functional. A plane-wave cutoff energy of 500 eV and a $4 \times 4 \times 4$ Γ -centered Monkhorst-Pack k mesh were employed. The lattice parameter of $a = 3.557$ Å is in good agreement with the experimental values $a = 3.567$ Å⁶⁴. A $20 \times 20 \times 20$ k-point grid was then used for the THERMACOND run to calculate the phonon properties, linewidth, and thermal transport.

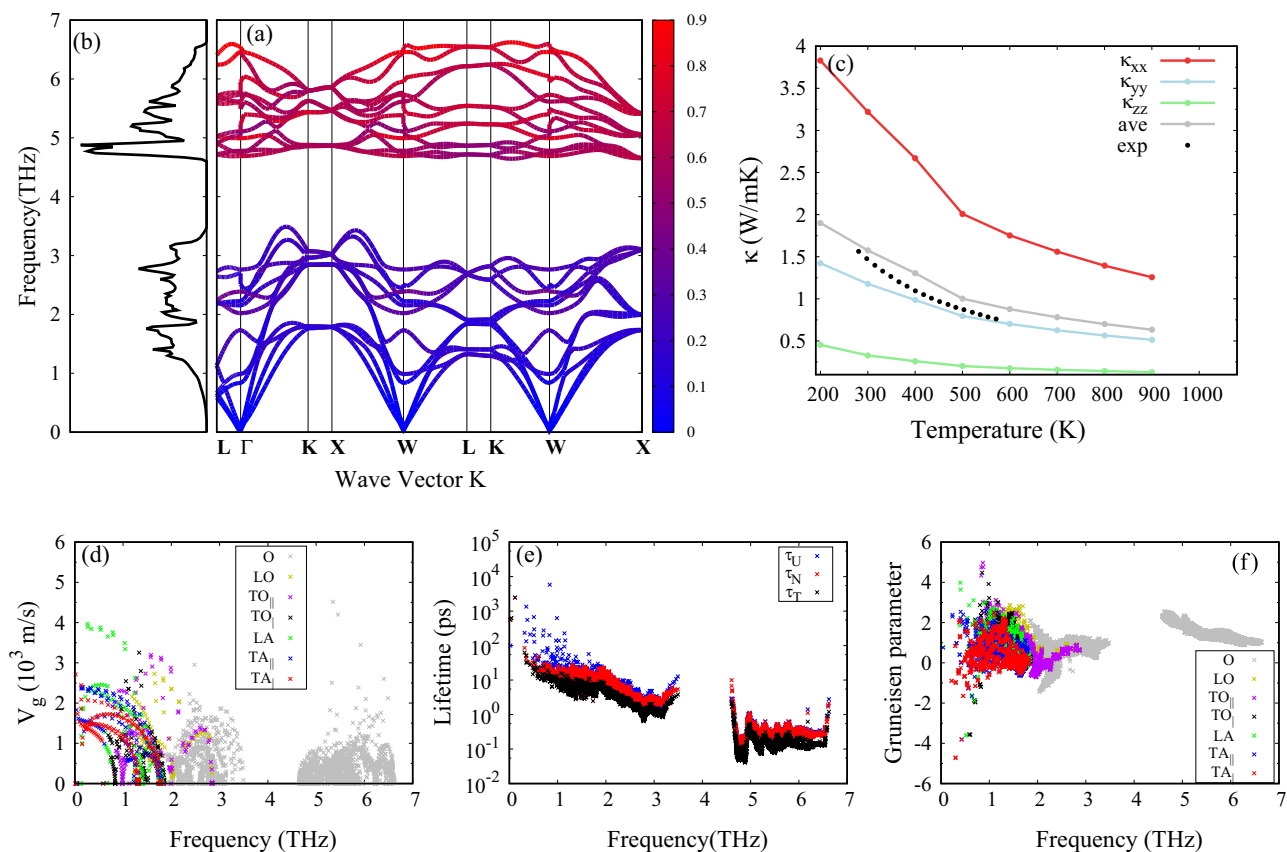


Fig. 3 | Phonon and phonon thermal transport properties of GeSe. a Phonon dispersion of GeSe used for thermal transport calculations. The color bar shows the intrinsic phonon linewidth in THz. **b** Phonon density of states. **c** Thermal conductivity as a function of temperature, compared with experimental data from

ref. 58. **d** Group velocity of each phonon mode versus frequency. **e** Intrinsic phonon-phonon scattering rates (black), with normal scatterings (red) and Umklapp scatterings (blue) separated. **f** Gruneisen parameters of each phonon mode in the irreducible zone as a function of phonon frequency.

The phonon dispersion along high-symmetry directions and the corresponding phonon density of states are shown in Fig. 4a, b, respectively. Each phonon branch is represented by its intrinsic linewidth at 300 K. These linewidths were computed along high-symmetry directions by integrating the imaginary part of the three-phonon self-energy over the same $20 \times 20 \times 20$ k-point grid. As shown in Fig. 4a, the optical modes exhibit higher scattering rates, as indicated by their broader linewidths. In contrast, the acoustic phonon modes exhibit longer lifetimes, as obtained from the RTA approximation (Fig. 4f) and the direct approach (Fig. 4g). These modes have the highest group velocities (Fig. 4e), playing a dominant role in thermal transport.

Subsequently, the thermal conductivity was computed by directly solving the phonon Boltzmann transport equation. Figure 4d presents the lattice thermal conductivity as a function of temperature and compares it with previously reported theoretical⁶⁵ and experimental values^{60–62}, as well as results obtained from ShengBTE²⁵. The ShengBTE results were obtained using the same displaced supercells, DFT calculations, and k-mesh. As shown in Fig. 4d, the thermal conductivity obtained within the RTA approximation and the direct approach agrees well with the previously reported theoretical and experimental results, as well as the ShengBTE results within both the RTA approximation and full iterative approach. This agreement validates the methodologies used in this work and confirms the reliability of the obtained results.

Figure 4f illustrate the frequency dependence of the phonon lifetimes, separated into total (black), normal (red), and Umklapp (blue) contributions, as obtained from the RTA approximation. In the RTA case (Fig. 4f), normal scatterings (red), which conserve total momentum and do not directly contribute to thermal resistance, dominates at lower frequencies,

whereas Umklapp processes (blue), which transfer phonon momentum outside the first Brillouin zone and lead to thermal resistance, become more significant at higher frequencies, leading to shorter phonon lifetimes. The total lifetime (black) follows the trend determined by the competition between these two scattering processes. However, the RTA underestimates the role of normal processes in thermal transport, as it treats normal scattering as a thermal resistance rather than a contributing factor to thermal transport. This limitation often leads to an underestimation of thermal conductivity.

Figure 4g compares the total phonon lifetimes obtained from the RTA (black) and direct (red) approaches. The direct approach yields longer phonon lifetimes, particularly in the low-to-mid frequency range, reinforcing the role of normal scattering in sustaining high thermal conductivity.

The comparison between these two methods highlights the limitations of the RTA in fully capturing normal phonon-phonon interactions. While Umklapp scattering remains the primary process limiting thermal transport, the direct solution shows that normal scattering enhances phonon lifetimes and leads to higher thermal conductivity compared to the RTA approximation. This confirms the necessity of solving the full phonon Boltzmann transport equation beyond the RTA for an accurate description of thermal transport.

Figure 5a, b compare the total phonon lifetimes obtained using the RTA and direct approaches with the results from ShengBTE. The agreement between these values further validates the reliability of the results.

Performance. Table 1 presents the time spent on the different computationally demanding sections of the code over various k-point grids for Germanium. These calculations were performed using Intel(R)

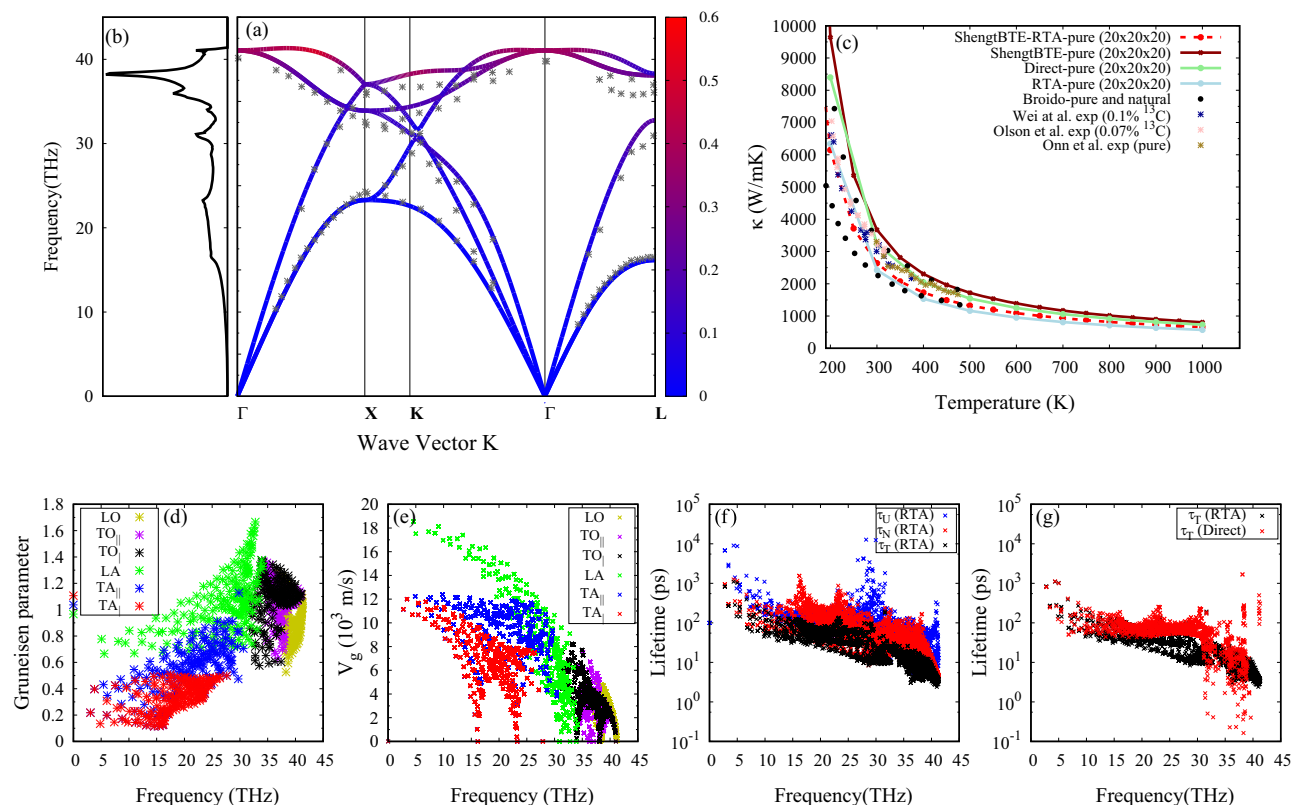
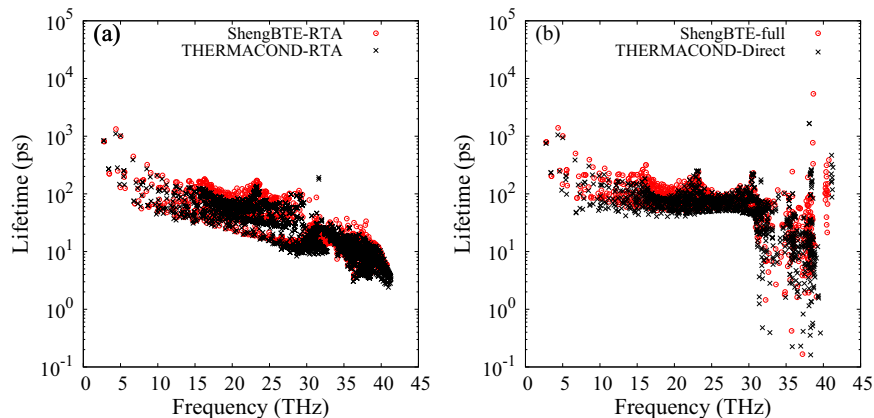


Fig. 4 | Phonon and phonon thermal transport properties of diamond.

a Diamond phonon dispersion with the color bar representing the intrinsic phonon linewidth in THz. Gray dots correspond to experimental data⁶⁴. **b** Phonon density of states. **c** Thermal conductivity as a function of temperature, compared with reported theoretical⁶⁵ and experimental results^{60–62}, as well as ShengBTE calculations. **d** Mode

Grüneisen parameters in the Brillouin zone as a function of frequency. **e** Group velocity of each phonon mode as a function of frequency. **f, g** Intrinsic three-phonon scattering lifetime (black), separated into normal (red) and Umklapp (blue) scatterings, within the RTA and direct approach, respectively.

Fig. 5 | Comparison of intrinsic three-phonon scattering lifetimes from THERMACOND and ShengBTE. **a** Intrinsic three-phonon scattering lifetimes (black) calculated using the THERMACOND code within the relaxation time approximation, compared with results obtained using the ShengBTE package (red). **b** Intrinsic phonon lifetimes computed using the direct approach in THERMACOND (black) are compared with ShengBTE's full iterative solution (red). Lifetimes are plotted as a function of phonon frequency.



Xeon(R) Gold 6248 CPU @ 2.50 GHz processors. As shown, the time taken to compute $V_3 \times \delta$ per k_{IBZ} per k_{FBZ} remains constant across different k -point grids. The total runtime is proportional to the time required per k_{IBZ} multiplied by the number of k_{IBZ} . It should be noted that the runtime depends on factors such as disk read/write speed, CPU performance, number of atoms, crystal structure, and the k -point grid. Additionally, once the three-phonon matrix elements $V_3 \times \delta$ are computed, they can be reused during the solution of the BTEs at different temperatures. The most time-consuming part of the code is the calculation of three-phonon matrix elements, which is linear in $N_{IBZ} \times N_{FBZ} \times N_{BANDS}^3 \times N_{FC3}$. All summations over the irreducible

zone are parallelized, so that scaling is linear in the inverse of number of CPUs. The direct solution using SVD or conjugate gradients minimization is relatively quick given that the collision matrix is $(N_{IBZ} \times 3 \times N_{BANDS}) \times (N_{IBZ} \times 3 \times N_{BANDS})$.

To assess the efficiency and accuracy of THERMACOND, Table 2 highlights key distinctions in solution methodologies, energy conservation techniques, and symmetry utilization compared to other widely used computational packages for solving the phonon Boltzmann transport equation. Unlike many existing codes, THERMACOND employs the tetrahedron method for Brillouin zone integrations, ensuring precise treatment of the energy conservation delta function. By leveraging crystal symmetry, it

Table 1 | Breakdown of the computational time (in CPU-hours) required for the various computationally intensive parts of the code over different k-point grids (first line is the number of kpoints in the full zone and second line is the corresponding number in the irreducible zone) for Germanium at T = 300 K, along with the thermal conductivities computed using the k-points within both the RTA and direct approaches

NFBZ NIBZ	4 × 4 × 4 11	8 × 8 × 8 50	10 × 10 × 10 85	14 × 14 × 14 196	16 × 16 × 16 276	20 × 20 × 20 495	30 × 30 × 30 1480
$V_3\delta/k_{IBZ}$ (Eq. (11))	2×10^{-3}	0.14	0.28	0.76	1.14	2.33	7.45
Phonon linewidth/ k_{bs} (Eq. (6))	0.07	0.46	0.90	2.67	3.82	7.85	27.30
Collision matrix / k_{IBZ} (Eq. (20))	2×10^{-6}	5×10^{-5}	8.14×10^{-4}	7.3×10^{-4}	16×10^{-4}	6×10^{-3}	0.17
$\kappa_{RTA}(W/mK)$	17.39	46.97	50.18	55.30	57.40	59.77	62.10
$\kappa(W/mK)$	17.61	49.29	52.07	57.70	59.24	62.31	64.70

Table 2 | Features and capabilities of various computational packages for predicting phonon properties and thermal transport

Feature/ Package	ShengBTE ²⁵	almaBTE ²¹	kALDo ⁶⁹	Phono3py ²⁴	ALAMODE ²⁰	Phoebe ³⁰	THERMACOND
BTE solution method	RTA, Iterative	RTA, Iterative, Direct Variance-reduced Monte Carlo	RTA, Iterative, Quasi-harmonic Green-Kubo (QHKG) for non-crystalline solids	RTA, Direct	RTA, iterative	RTA, Variational, Iterative, Relaxons, Direct	RTA, Iterative, Direct
Harmonic force constants	DFPT, Finite difference	Finite difference using phonopy	Finite difference (DFT or MD) and DFPT, from QE, Phonopy, LAMMPS, GROMACS, hiPhive	Finite difference, DFPT	Linear least-squares fitting	DFPT, Finite difference using phonopy/phono3py and Quantum ESPRESSO (QE)	FOCEX
Cubic force constants	Finite difference, Symmetry reduction	Finite difference using ShengBTE	Finite difference (DFT or MD) and DFPT, from QE, Phonopy, LAMMPS, GROMACS, hiPhive	Finite difference	Linear least-squares fitting	Finite difference using phono3py and ShengBTE	FOCEX
Quartic force constants	—	—	—	—	—	—	Yes using FOCEX
Energy conservation δ – function	Gaussian	Adaptive Gaussian	Gaussian, Lorentzian, triangular	Tetrahedron	Lorentzian, Tetrahedron	Adaptive Gaussian Tetrahedron (only for DoS)	Lorentzian, Tetrahedron
Crystal symmetry	Reduces distinct force constants and optimizes Brillouin zone integration	Reduces distinct force constants and optimizes Brillouin zone integration	Scattering selection rules (Dirac delta function), Reduces distinct force constants using Phonopy, hiPhive	Reduces distinct force constants and optimizes Brillouin zone integration	Reduces distinct force constants	Reduces distinct force constants and optimizes Brillouin zone integration	Reduces distinct force constants and optimizes Brillouin zone integration
Phonon dispersion	—	—	Yes	Yes	Yes	Yes	Yes
Phonon linewidth	Scattering rates	Scattering rates	Scattering rates	Imaginary parts of self-energies using Tetrahedron	Imaginary parts of self-energies	Scattering rates	Imaginary parts of self-energies
Thermal conductivity	Yes	Yes	Yes (both via BTE and QHKG)	Yes	Yes	Yes	Yes

optimizes computational efficiency by restricting calculations to the irreducible wedge of the Brillouin zone (IBZ), significantly reducing memory usage and computational cost. THERMACOND supports both iterative and direct (non-iterative) solutions, with the direct approach eliminating numerical instabilities associated with iterative schemes, thereby accelerating convergence. Parallelized using MPI, the code efficiently handles large k-point meshes, making it well-suited for high-throughput calculations. The combination of crystal symmetry-based reduction, tetrahedron-based energy-conserving integration, MPI parallelization, and the direct solution method contributes to faster convergence and enhanced accuracy, validating THERMACOND's robustness for phonon thermal transport calculations.

Methods

The calculation of lattice thermal conductivity and phonon scattering rates using THERMACOND involves reading interatomic force constants, computing phonon frequencies and group velocities, phonon transition probabilities, and finally solving the linearized phonon Boltzmann transport equation. Following the workflow presented in Fig. 1, both harmonic and

anharmonic interatomic force constants are obtained utilizing the direct supercell approach as implemented in FOCEX, part of the Anharmonic LATtice DYNamics (ALATDYN) suite of codes. In this work, the supercell DFT calculations were performed using VASP for Ge and diamond and SIESTA for GeSe. Detailed computational parameters of DFT calculations, including supercell sizes, neighbor shells, cutoffs, and k-point grids, are provided in the previous sections for Ge, GeSe, and diamond individually. Subsequently, after obtaining second-order harmonic and third-order anharmonic IFCs using FOCEX, the thermal conductivities of Ge, GeSe, and diamond were computed using a converged k-point grid, as detailed in the previous section, by solving the phonon Boltzmann transport equation using the direct approach as implemented in THERMACOND. This code is written in Fortran90, and is built with MPI support using OpenMPI. The compilation process typically involves loading the necessary modules (gcc and openmpi) to ensure compatibility with high-performance computing environments. While the iterative method is also implemented, it is recommended to run THERMACOND using the direct approach to have accurate results and avoid convergence issues. The latter employs either

singular value decomposition (SVD) or conjugate gradient (CG) minimization, both of which yield comparable computational performance and identical results. The execution of THERMACOND is controlled via the <latdyn.params> input file, which allows users to specify key parameters, including k-point mesh size, smearing method, temperature range, parallelization (MPI), isotope scattering, and solver type (iterative vs. direct, SVD vs. CG). Upon execution, THERMACOND generates standardized output files that contain all relevant phonon thermal transport data, making it easy for users to analyze the results. Additionally, to streamline post-processing, gnuplot script files are included with the package, allowing users to effortlessly generate colorful plots of phonon thermal transport properties for studied examples.

Data availability

The input files required to run THERMACOND and reproduce the results presented in this work are available on GitHub⁶⁶, and at the University of Virginia Dataverse⁶⁷.

Code availability

The THERMACOND codes, and more generally the Anharmonic LATtice DYNamics (ALATDYN) suites of codes may be accessed and downloaded from GitHub⁶⁶. Documentation is available online⁶⁸.

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Author contributions

K.E. conceived the project and wrote the phonon properties, V_3 calculation and the relaxation time approximation (RTA) version of the code using the imaginary part of the self-energy as detailed in ref. 36. S.L. developed the iterative approach and implemented the tetrahedron method. S.N.S. incorporated mass disorder scattering rates into the collision matrix, symmetry reduction into the iterative approach, developed a direct solution approach, parallelized the code using MPI, tested the code for different materials and approaches, and validated the results by comparing them with calculations from alternative available codes, and wrote the manuscript. All authors analyzed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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